Exam Date & Time: 27-Dec-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

V SEMESTER B.TECH END SEMESTER MAKEUP EXAMINATION, DEC 2024 - JAN 2025

BASIC CLINICAL SCIENCE II [BME 3121]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions.

Instructions to Candidates:
[i] Missing data may be suitably assumed
[ii] No formula book will be provided

1A)	Outline the instrumentation of digital subtraction angiography.	(4)
1B)	List the steps involved in processing a Digital Radiography image.	(3)
1C)	Discuss the working principle of Image intensifier.	(3)
2A)	Discuss briefly the various generations of Computed Tomography.	(5)
2B)	Describe about scattering foil and flattening filter.	(3)
2C)	Describe the three methods of radiation control and their correlation with dose.	(2)
3A)	Define acute radiation syndrome, and describe the clinical manifestations with types of syndromes and phases.	(3)
3B)	Enumerate the 5 R of radiobiology.	(3)
3C)	What is Fundus fluorescein angiography (FFA)? Write the principle of FFA. Mention two clinical conditions in which FFA is useful.	(4)
4A)	Write 2 advantages of contact lenses over spectacles.	(4)
4B)	What is perimetry? What are the different perimeters used routinely?	(2)
4C)	Describe the different types of brain waves recorded in EEG. How do these waveforms vary with the state of consciousness?	(4)

5A)	Explain the underlying principle behind optogenetics. What are the primary steps involved in applying optogenetics for neuromodulation?	(4)
5B)	Describe the working principle of fNIRS and its limitations in neuroimaging. What type of neural activity can it best capture?	(3)
5C)	Explain how Nerve Conduction Studies (NCS) are used to diagnose Carpal Tunnel Syndrome (CTS).	(3)

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